

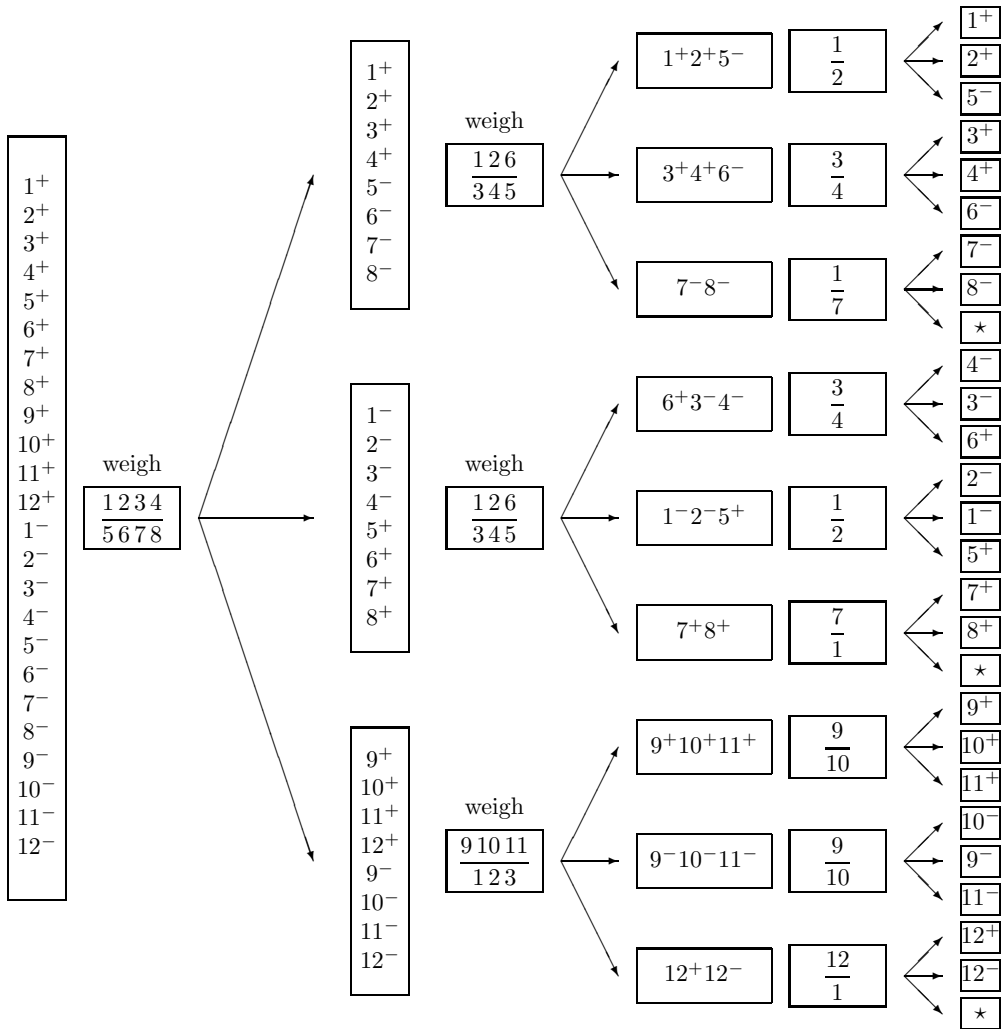


Figure 4.2. An optimal solution to the weighing problem. At each step there are two boxes: the left box shows which hypotheses are still possible; the right box shows the balls involved in the next weighing. The 24 hypotheses are written $1^+, \dots, 12^-$, with, e.g., 1^+ denoting that 1 is the odd ball and it is heavy. Weighings are written by listing the names of the balls on the two pans, separated by a line; for example, in the first weighing, balls 1, 2, 3, and 4 are put on the left-hand side and 5, 6, 7, and 8 on the right. In each triplet of arrows the upper arrow leads to the situation when the left side is heavier, the middle arrow to the situation when the right side is heavier, and the lower arrow to the situation when the outcome is balanced. The three points labelled $\star$ correspond to impossible outcomes.